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ORIGINAL RESEARCH

Waging warfare against states: the deployment of artificial intelligence in cyber espionage

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Abstract
Cyber espionage has significantly been viewed as a risk towards nation-states, especially in the area of security and protection of Critical National Infrastructures. The race against digitisation has also raised concerns about how emerging technologies are defining how cyber activities are linked to waging warfare between States. Real-world crimes have since found a place in cyberspace, and with high connectivity, has exposed various actors to various risks and vulnerabilities, including cyber espionage. Cyber espionage has always been a national security issue as it does not only target States but also affects public–private networks, corporations and individuals. The challenge of crimes committed within the cyber realm is how the nature of cybercrimes distorts the dichotomy of state responsibility in responding to cyber threats and vulnerabilities. Furthermore, the veil of anonymity and emerging technologies such as artificial intelligence have further provided opportunities for a larger scale impact on the state for such crime. The imminent threat of cyber espionage is impacting the economic and political interactions between nation-states and changing the nature of modern conflict. Due to these implications, this paper will discuss the current legal landscape governing cyber espionage and the impact of the use of artificial intelligence in the commission of such crimes.

Keywords Cyber espionage · Artificial intelligence · Cyber operations · Warfare · International law

1 Introduction

Cyber espionage is coined as a type of cyber warfare that includes disruptions, surveillance, damage, and deletion of information. Cyber espionage threats are powerful enough to immobilise a state and disrupt the running of critical national infrastructures, where the sabotage of one sector may result in total system failure, data leakage, and even system harm. If one sector fails, it would cause a domino effect on other critical sectors which could result in a national catastrophe. The European Union Agency for Cybersecurity (ENISA) defines it as a type of threat committed by using a computer network to gain illicit access to confidential information usually held by state governments or organisations [1]. Such information or intelligence refers to information which is of military, political and economic value. The United States Central Intelligence Agency (CIA) contended that intelligence involves knowledge and foreknowledge of the world which drives decisions by policymakers. The Tallinn Manual, which is one of the first documents that outlines the rules for nation-state cyber warfare has stated that espionage as an act undertaken clandestinely or under false pretences that uses cyber capabilities to gather or attempt to gather information [2].

Previously, espionage was usually done by states sending their most skilled agent into the territory of their physical adversaries. The development of technology has changed the way espionage is conducted where states now spy on their enemies by using more sophisticated methods such as drones and spy balloons. For years, state and non-state actors have been engaging in spying to gain information, assess potential threats, and prepare for attacks [3]. Essentially, throughout history, states have continued to engage in clandestine activities against each other through spying and sabotaging [4]. This type of practice has long been viewed as an attribute of warfare, state intelligence and combatant engagement. The race towards digitalisation and the emergence of disruptive technologies are now viewed to be an

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expansion of the term cyber warfare. This also meant that nation-states began to move their focus to the protection of sectors within the critical national infrastructure which required special cyber protection and security.

In combination with the growing sophistication of cyber actors and activities, it further expands opportunities for coordinated and advanced attacks that can speedily disrupt the operational efficiency of designated critical national infrastructure e.g., the operation of the financial markets, electricity, security infrastructure and general elections [5]. When cyber espionage is part of a broader military campaign it can also lead to disruption of public infrastructure and property [6]. Cyber espionage attacks can be motivated by monetary value or may also be deployed in conjunction with military operations, or as an act of cyber terrorism or cyber warfare.

Espionage has become an imminent threat and risk to nations, especially in understanding how technology shares and influences the relationship between states. Many nations and international bodies around the world have created their own definition of cyber espionage and it has been proven to be difficult to reach a consensus due to the secretive nature of espionage and its relation to warfare. On the other hand, states have also argued that cyber espionage tactics may be used by intelligence services to protect national security by preventing terrorist attacks [7]. It can also be used in war where the information collected from the enemy can be used successfully to gain warfare advantage [8]. With the realization of the opportunities that cyberspace offers, governments across the world are building on cyber-forces, which are tasked with accomplishing their national cybersecurity policy goals and intelligence agendas [9].

Against this backdrop, the implications for international security are serious and far-reaching. As nation-states contend for dominance in cyberspace and geopolitical tensions escalate, the risk of Artificial Intelligence (AI) -enabled cyber activities will unarguably exacerbate tensions. The potential for misattributed attacks, unintended escalation, and the increase of cyber operations threatens to destabilise the balance of power and heighten the risks of conflict escalation [10]. It is imperative to therefore identify the emergent contours of warfare in the digital age and develop sound strategies for safeguarding international security.

2 International law and the waging of wars between states

Espionage has always been a feature of interstate relations deeply rooted in secret intelligence-gathering activities conducted between states. Article 2 of the 1945 Charter of the United Nations (UN Charter) explicitly explained that all member states should refrain from using force or threats that encroach on another state's territorial integrity or political independence. Despite the availability of an international framework for the governance of espionage, specific treaties and conventions addressing the intrusion are notably absent. The commission of espionage however is dealt with under principles of international humanitarian law, sovereignty and customary practices.

The concept of espionage also violates the duty of non-intervention which was also mandated under the UN Charter. Jamnejad and Wood [11] explains that the principle of non-intervention includes the right to sovereignty, territorial integrity and political independence of every State. The principle has also been accepted to apply to cyber operations. However, the non-intervention principle only applies to States as it is an inter-state doctrine. Any activities conducted by non-state actors do not apply unless through the doctrine of attribution in international law. Within the context of state-sponsored cyber operations, which target individuals and companies, the non-intervention principle does not apply as it does not affect the state's exclusive rights to sovereign functions [12]. To date, in so far cyber operations such as cyber espionage, states have only responded to attacks violating the states' ability to exercise their sovereign powers [12]. This can be seen by the response of the Not-Petya attack targeting Ukraine's critical national infrastructures which was labelled as an act disregarding Ukrainian sovereignty as appose to the spear fishing attack campaign aimed at private universities, companies and NGOs by Iran which was not to be said a violation of international law [13].

The principle of non-intervention and its requirement of fulfilling the element of violation of sovereignty proves challenging as the nature of cyberspace which allows covert, anonymous and remote operations which can also go beyond borders. Furthermore, careful investigations and cooperation between international partners are crucial in order to track and find the perpetrators. Recent attacks have shown more coercive and intrusive patterns where these intrusions often target the critical national infrastructure of States such as transportation, energy and manufacturing [13]. However, Oorsprong [14] highlighted that espionage could hardly be considered as an international crime as underlying reasons for collecting the information would usually not constitute to interference or violate the non-intervention principle.

Regardless of the argument of how cyber operations such as cyber espionage would go against international law principles, the majority of states have always viewed espionage-like acts to be an important element in maintaining peace, stability and security between nation-states. Interestingly, Hernandez [15] and Scott [16] have also highlighted that the practice of espionage between states is legal on the basis

of anticipatory or pre-emptive self-defence. Such belief is derived from the power of knowing the ability and strength of other states can be determined, and the balance of power and status quo can be achieved which results in non-confrontational behaviour [17].

Apart from nation-states engaging in espionage, non-state actors also have the ability to take part in espionage activities. However, nation-states will always play a central role in conflicts among states due to the financial, technological and cyber capabilities to conduct sophisticated operations and share geopolitical dynamics in cyberspace. Rauta [17] contends that these proxy groups and non-state actors engage in warfare through sponsorship of the state to prevent attribution. These entities would carry out covert activities, data mining, and cyber-attacks to advance the strategic interest on behalf of the sponsoring state. It must be noted that actions or behaviour which could be triggering an act of war would be instances where unfriendly economic or commercial actions or any behaviour that insults national price. However, acts of cyber espionage are stealthier rather than traditional warfare which is more transparent. This in turn blurs the line of retaliation and response due to the targets are usually private companies and not directly towards the State.

In another sphere, the dichotomy between developed and developing nations adds another division to the conversation. Spectres of superiority and dominance between states have contributed to the ongoing geopolitical and cyber-diplomatic struggles globally. Superiority is being determined by one nation's economic, political and military strength which has led to conversations and debates on cyber governance [18]. States such as China and Russia have resorted to diverse covert activities to gain economic and technological power. The advancement of technology has broken traditional barriers, and the cyberspace environment now offers states the opportunity to take advantage of such cyber capabilities [19]. This has inherently sparked the need for a unified framework to govern cybercrime in the form of the world's first international cybercrime treaty [20].

3 Proliferation of artificial intelligence (AI) in cyber espionage

The convergence of law and technologies has brought about various implications and challenges. The use of emerging technologies afforded by AI has proven to be the holy grail in combatting crimes within the real world and cyberspace. It also has become an integral part of society, driving innovations across sectors including legal or illegal cyber operations. The deployment of AI in the fight against cybercrime has also facilitated the investigation of crimes committed in cyberspace, especially for entities that hide behind the shadows of cyberspace and their digital footprints. The Swiss Federal Intelligence Service (FIS) Annual Situation Report (2023) highlighted that espionage is driven by various motives which involve various aims such as economic and political superiority. The FIS also contended that the availability of new emerging technologies acts as a sandbox for various actors to develop new solutions to problems including the need for states to focus on lingering military conflict [21].

The use of AI aids in detecting cyber espionage including the use of biometric recognition technologies such as facial detection to sift through data available online and provide immediate notification [22]. The emergence of Generative AI (GenAI) has completely changed the landscape of commission and protection against cyber espionage. GenAI has revolutionised tools and solutions used by States to safeguard their systems from infiltration [23]. Such technology provides defence against automation, cybersecurity reporting, threat intelligence, identification of attacks, incidence response and malware detection. Apart from that, GenAI has also unleashed both known and unknown cyberattack vectors in the form of automated hacking, code-adapting malware and AI-crafted phishing e-mails.

The increased dependency on AI has provided law enforcement agencies with the tools to fight cybercrime more effectively. New emerging technologies such as build-in cyber espionage search engines which eliminates cumbersome elements of extracting information-laden threats and discovering digital footprints of offenders in order to link the relevant information which is time-consuming [23]. The use of such technology has broken barriers of time-consuming work with the capability of shifting through unlimited volumes of data, the least amount of time and minimal resources to track down leads for any vulnerabilities and threats against the State [23]. This has also led to the eradication of threats of violence and improve security measures within the State.

The duality impact of AI must also be recognised. AI has proven to be lethal, especially in coordinating cyber-attacks via sophisticated algorithms that can penetrate even the most complex security systems, data theft of sensitive information such as state secrets or military capability information, and disruption of multisector services [24]. Samuel-Okon et al. [25] contend that AI-driven malware can evade traditional detection methods and cause extensive damage as it has the capability to adapt and evolve causing widespread data breaches. AI algorithms can analyse a vast amount of data derived from various sources such as public records, network traffic and social media in order to identify potential targets. These algorithms can also develop stealthy tools to infiltrate and bypass security defences and

intrusion detection systems. Moisset [26] contends that AI algorithms can assist in prioritising targets based on factors such as potential intelligence value, susceptibility to exploitation and significance. Catalano et al. [27] highlighted that adversarial machine learning also enables attackers to evade signature-based detection.

AI is no longer confined to science fiction but is now a critical component of military strategy. Nations are actively integrating AI into their defense capabilities, leading to a paradigm shift in how wars are fought. In recent years, characterized by rapid technological advancement, the emergence of AI and cyber espionage has become a defining feature of modern warfare and international security [28]. As nations increasingly harness the power of AI to bolster their defense capabilities and conduct intelligence operations, the landscape of conflict has evolved to another dimension—one where states use various means to try and gather information from other states without consent [28].

The emergence of GenAI has also changed the landscape of the commission and protection against cyber espionage. GenAI-aided attacks have now unleashed both known and unknown cyberattack vectors. Gupta *et. al* highlighted that GenAI acts as a double-edged sword in cybersecurity as it facilitates and benefits both sides of the coin [28]. Nation states have now moved to GenAI tools to safeguard their systems from infiltration. Information gathered from Large Language Models (LLM) trains intelligence data such as risk, vulnerabilities and prediction of attacks [28]. States now have the capability to process large information to enhance their intelligence response and even predict emerging and future threats [29]. Furthermore, such advanced technology is also important in developing effective ethical guidelines for strengthening cyber defense within a system [30]. According to Gupta, the use of GenAI provides support in cyber defense automation, cybersecurity reporting, threat intelligence, identification of cyber-attacks, incident response guidance and even malware detection [30].

The worldwide movement to govern the use of AI has also impacted the way the State responds to the use of GenAI in the commission of cyber espionage. The legal guidelines and practices such as the EU AI Act aim to give clear guidelines on the misuse and responsibility of users when using such technology. Such moves have leveraged the use of such technology and have also pushed states to give clear guidelines on the use of such emerging technology.

4 The legal response to cyber espionage

Effective governance of cybercrimes and cyber operations have been at the forefront of every nation, especially in eradicating national security threats. The Office of the National Counterintelligence Executive has also highlighted that cyber espionage heightens the risk to international peace and security [31]. The United Nations Special Rapporteur on Counterterrorism and Human Rights have also called for a legal response to spyware proliferation. However, it must be noted that the line between lawful and unlawful cyber espionage or intrusions remains unclear. With the current economic and political global landscape, states have now turned to engage in diverse cyber activities on the pretext of national security [32]. Apart from relying on traditional laws such as national legislation and statutes to govern cybercrimes, states have now moved to various other mechanisms in addressing cybersecurity matters such as devising cybersecurity strategies in the form of state policies in order to maintain cyber resilience across sectors [32]. A cybersecurity strategy uses a top-down approach which contains objectives and protocols in order to protect a state's critical national infrastructure. The strategy is designed to maximise the security and resilience of a nation by outlining the duties and obligations of major stakeholders within the state ecosystem [33]. States also view that having a strategy in place is a proactive approach to mitigate the risks and threats of cyber-attacks. Furthermore, such strategies also recognises the ever-evolving advancement of technologies and continuously devise ways for the state to adapt and align with national security.

The Assange and Snowden leak has revealed the gravity of intrusion and data breach to states which motivates them to seek a harmonised approach to regulate and mitigate the risk of cyber espionage. States have now moved towards the curtailing of remote access cyber espionage by deploying state-of-the-art technologies to restrict such operations. In 2023, aware of the risk on infiltration and the risk of espionage to national security, the United States passed the Presidential Initiative for Democratic Renewal which intends to control the misuse of technology and digital authoritarianism [32]. The Initiative also aims at combatting digital domination which includes an executive order to prohibit the use of spyware which poses a risk to national security. Apart from that, the United States also enacted the Espionage Act which restricts the access and disclosure of information which can be used against the United States or provide undue advantage to a foreign state which would compromise national security [34]. The former United States National Security Agency contractor, Edward Snowden was charged under this Act, but has yet to face trial due to being granted the right of asylum in Russia.

Traditional legislation governing cyber espionage can be found in every country in one form or another. In China, cyber espionage is governed under the National Intelligence Law 2017 which aims to strengthen and safeguard national intelligence work and preserve state security and interest

[35]. Interestingly, the law also attempts to balance the securitization of sectors to the protection of human rights and the interest of individual citizens. In Russia, the Criminal Code of the Federation, Law NO. FZ-190 criminalise espionage and it is seen as treason. The law includes acts of assisting foreign states, disclosure of national secrets or anything that would be detrimental to the security of the Russian Federation [36].

The European Union has also taken steps to regulate cyber espionage when recently the European Parliament called for a stricter approach to the use and abuse of spyware. Furthermore, in February 2024, the United Kingdom and France, hosted nation states, regional organizations and other stakeholders in an inaugural conference to tackle the proliferation and irresponsible use of commercial cyber intrusion tools and services [37]. The conference also led to the signing of the cyber declaration—The Pall Mall Process: Tackling the Proliferation and Irresponsible Use of Commercial Cyber Intrusion Capabilities [38].

The Dutch Military Intelligence and Security Service (MIVD) highlighted that Western governments, defence companies and international organisations are now being targeted with cyber espionage [39]. MIVD also contended within it 2024 annual report that Chinese spies has recently targeted Dutch semiconductor, aerospace and maritime industries in order to gather intelligence in strengthening its armed forces. However, China have always routinely denied any form of allegations relating it to cyber espionage [40].

The first quarter of the year has also seen Western governments especially within Europe being under siege by cyberattacks and operations of espionage. The United Kingdom Ministry of Defence has also suffered cyberattacks which exposed the sensitive data of thousands of its troops [40]. The breach targeted the weakness of a third-party contractor who runs the payroll systems for the Ministry which leaked more than 270,000 data of former and current armed forces members. Germany was also targeted when a 38-min audio recording of four top military personnel discussing the support for Ukraine including using missiles to attack Russia [40].

The legal response to cyber espionage takes many forms and challenges. Apart from what has been discussed above, such attacks depend on the type of cyber operations and whether the implications are confined to the national level or have impacted internationally. For example, an Iranian-based group named Helix Kitten has been notorious for targeting companies or organisations within the sectors of transportation, finance, hospitality and telecommunications and using sophisticated methods via structured spear-phishing attacks to targeted personnel. Another national state actor, which is an organisation that is supported by a specific nation is a group labelled as Fancy Bear, a Russian-based group that is linked to the Russian Intelligence Service. Their primary aim is to target U.S. political organisations and European military organisations through zero-day vulnerabilities and malware drop websites disguised as new sources.

The response to such invasive actions can be both military and non-military. Non-military responses would include diplomatic measures such as demarches and expelling diplomats, legal actions such as cybercrime legislation and international legal cooperation, cyber defence and counterintelligence and also economic sanctions. However, the unequal level of digital capacity between nation-states also serves as a barrier to efforts of harmonisation of the law and better governance of cyber espionage. The challenges differ across regional boundaries considering limited resources, technical capabilities, and other social realities. A general lack of awareness of cybersecurity best practices also leaves many states vulnerable.

5 International security and cyber espionage governance

The ever-evolving threat of cyber espionage cannot be denied especially within the realm of cybersecurity with implications that could range from national security to political and economic stability. This type of cyber operation has been a common practice by states and non-state actors who practice recognisance activities to measure the strength and capability of other adversaries. From using old tactics such as sending agents into other jurisdictions to gather intelligence to the emergence of new technologies which transcend the recognition of borders and time, cyber espionage has time and time again proven to be a risk that needs to be managed.

Calls for criminalisation and clear classification of this cyber operation have led to efforts to harmonise international perspectives due to domestic laws being insufficient to address the challenges to national security, privacy and data protection and extraterritorial jurisdiction. The apparent global digital divide and access to justice for all must also be addressed. The best way forward would be to have a treaty to govern and clearly outline the use of emerging technology in the commission of cyber espionage. However, efforts to materialise an international treaty to govern such cybercrime pose a multitude of challenges due to the different interests and priorities of states especially in the area of sovereignty and security. Scholars have highlighted that a more realistic approach would be to completely prohibit cyber espionage and have in place norms and practices agreed upon by all states. Greiman (2018) further states that if States want non-binding responsibilities in cyberspace,

therefore their actions and practice must demonstrate their willingness to achieve that goal.

The increasing reliance on artificial AI by state intelligence would depend on the capability and operationalisation of intelligence activities. States will invest and focus on advancing their technological resources which will enhance their cyber power and international presence. This means that in the coming years, there will be an urgent need to nation states, international organizations and other stakeholders to consider a normative framework for the use of such capabilities. Furthermore, AI governance is vital as a way forward to invest in the development of AI resources. The passing of the AI Act has provided an opportunity for other states to also initiate their own strategies, policies and legal framework in governing AI and other emerging technologies. Apart from finding ways to regulate technologies, enhancing cybersecurity measures and deterring emergent threats are also critical in ensuring effective governance of cyber espionage. Collaborative intelligence is also essential to identify emerging risks and threats more effectively. Finally, by fostering cybersecurity awareness, investment in state-of-the-art defensive technologies and strengthening international cooperation, states can mitigate the risks of cyber espionage and create a more secure and trusted digital ecosystem.

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